# PLANILLA DE: LOSAS MACIZAS

$\sigma'_{bk}$ = 170 Kg / cm2
$\sigma_{ek}$ = 4200 Kg / cm2

| NIVEL | DESIG. | Luz m lx / ly | ε=ly/lx | Cargas T/m2 g | q | M(+) Tm Mx | My | Dimens. Esp. d | Armadura (por metro) cm2 | Ø (mm) | Sep. (cm) | Caballete 1/2 de Fe bl cm | al cm | ad cm | bd cm | Armad. de Repart. Ø (mm) | Sep. (cm) | observaciones |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (0/1) | LE (0/1) | 4.00 | ---- | 0.885 | 0.885 | 1.77 | 12.5 | 14.0 | 7.8 | 10 | 10.0 | 15 | 55 | 55 | 15 | 6 | 20 | (DIR "Y") "Losas Escaleras" |
| ① | L 102 | 2.65 / 2.65 | 1.00 | 0.800 | 0.400 / 0.400 | 0.30 / 0.30 | 7.0 / 6.5 | 8.5 | 2.44 / 2.44 | 6 / 6 | 11.5 / 11.5 | 15 / 45 | 40 / 40 | 40 / 40 | 15 / 20 | ---- | ---- | (DIR "X" e "Y") "Losas Piso Sanitarios" |
| ①' | L 02 | 2.65 / 2.65 | 1.00 | 0.560 | 0.280 / 0.280 | 0.17 / 0.17 | 7.0 / 6.5 | 8.5 | 2.15 / 2.15 | 6 / 6 | 13.0 / 13.0 | 15 / 15 | 40 / 40 | 40 / 40 | 15 / 15 | ---- | ---- | (DIR "X" e "Y") "Losas Techo Sanitarios" |
| | L 03 | 1.85 / 2.65 | 1.40 | 0.560 | 0.41 / 0.15 | 0.14 / 0.12 | 7.0 / 6.5 | 8.5 | 2.15 / 2.15 | 6 / 6 | 13.0 / 13.0 | 15 / 15 | 30 / 40 | 30 / 40 | 15 / 15 | ---- | ---- | (DIR "X" e "Y") "Losas Techo Sanitarios" |
| ② | L PTR | 2.65 | ---- | 1.1 | 1.1 | 0.97 | 9.5 | 11.0 | 5.0 | 8 | 10.0 | 15 | 40 | 40 | 15 | 6 | 20 | (DIR "X") "Losas Piso T. Reserva" |

# PLANILLA DE: LOSAS NERVURADAS (ALIVIANADAS)

$\sigma'_{bk}$ = 170 Kg / cm2
$\sigma_{ek}$ = 4.200 Kg / cm2

MOLONES DE POLIEST. EXPANDIDO

| NIVEL | DESIG. | Luz m | ε | Cargas T/m2 | M(+) Tm | bo | do | h | b | d | Distancia a Ejes x | y | Fet (inf) cm2 | Nº | Ø | Fet' (sup) cm2 | Nº | Ø | Hierros Doblados A Nº | Ø | B Nº | Ø | Estribos Ø | Sep. | Armadura de Repart. Ø | Sep. | Tension de Corte τA | τB | observaciones |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ① | L 100 | VAR | ---- | 0.850 | 0.332 / 3.66 | 14.0 | 31.0 | 29.5 | 39.0 | 6.0 | ---- | 39.0 | 5.6 | 5 | 12 | 0.28 | 2 | 4.2 | 2 | 12 | 2 | 12 | 4.2 | 25 | 4.2 | 16.5 | 3.78 | 3.78 | (Dir. "Y") "MOLON DE POLIEST. EXP." |
| | L 101 | VAR | ---- | 0.850 | 0.332 / 3.66 | 14.0 | 31.0 | 29.5 | 39.0 | 6.0 | ---- | 39.0 | 5.6 | 5 | 12 | 0.28 | 2 | 4.2 | 2 | 12 | 2 | 12 | 4.2 | 25 | 4.2 | 16.5 | 3.78 | 3.78 | (Dir. "Y") "MOLON DE POLIEST. EXP." |
| ② | L 200 | VAR | ---- | 0.600 | 0.234 / 2.58 | 14.0 | 26.0 | 24.5 | 39.0 | 6.0 | ---- | 39.0 | 4.5 | 4 | 12 | 0.28 | 2 | 4.2 | 2 | 12 | 2 | 12 | 4.2 | 25 | 4.2 | 16.5 | 3.20 | 3.20 | (Dir. "Y") "MOLON DE POLIEST. EXP." |
| | L 201 | VAR | ---- | 0.600 | 0.384 / 0.90 | 14.0 | 26.0 | 24.5 | 64.0 | 6.0 | ---- | 64.0 | 2.4 | 3 | 10 | 0.28 | 2 | 4.2 | 1 | 10 | 1 | 10 | 4.2 | 25 | 4.2 | 16.5 | 2.40 | 2.40 | (Dir. "Y") "MOLON DE POLIEST. EXP." |
| | L 203 | VAR | ---- | 0.600 | 0.384 / 0.18 | 14.0 | 26.0 | 24.5 | 64.0 | 6.0 | ---- | 64.0 | 1.5 | 3 | 8 | 0.28 | 2 | 4.2 | 1 | 8 | 1 | 8 | 4.2 | 25 | 4.2 | 16.5 | 1.12 | 1.12 | (Dir. "Y") "MOLON DE POLIEST. EXP." |

# PLANILLA DE: VIGAS DE FUNDACION, (TENSORES Y RIOSTRAS)

$\sigma'_{bk}$ = 170 Kg / cm2 (Hº : COMPRESION)
$\sigma_b$ = 15 Kg / cm2 (Hº : TRACCION)
$\sigma_{ek}$ = 4200 Kg / cm2 (ACERO)

| Designacion | Luz m | RA tn | RB tn | F tn | bo | h | do | tramo Fes Nº | Ø | tramo Fei Nº | Ø | apoyo Fes Nº | Ø | apoyo Fei Nº | Ø | Adicional Fe ad Nº | Ø | Corte τA kg/cm2 | τB | Estribos Cerrados Ø | Sep. | observaciones |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| VT [6-1] | 2.40 | 5.00 | 5.00 | 10.0 | 20.0 | 35.0 | 37.5 | 3 | 12 | 3 | 12 | 3 | 12 | 3 | 12 | -- | -- | 13.33 | 13.33 | 6 | 18 | NIVEL SUPERIOR DE VIGAS (NIVEL -1) (=-0.35m); VIGAS TENSORES; NOTA: A 15% DE LUZ EN C/APOYO, DENSIFICACION DE ESTRIBOS CERRADOS EN (30%) |
| VT [6-7] | 8.72 | 5.00 | 5.00 | 10.0 | 15.0 | 42.5 | 45.0 | 3 | 12 | 3 | 12 | 3 | 12 | 3 | 12 | -- | -- | 14.8 | 14.8 | 6 | 18 | |
| VT [7-8] | 7.95 | 9.7 | 9.7 | 19.4 | 35.0 | 37.5 | 40.0 | 2/1 | 16/12 | 2/1 | 16/12 | 2/1 | 16/12 | 2/1 | 16/12 | -- | -- | 13.86 | 13.86 | 6 | 18 | |
| VT [9-10] | 8.35 | 7.4 | 7.4 | 14.8 | 27.5 | 37.5 | 40.0 | 2 | 16 | 2 | 16 | 2 | 16 | 2 | 16 | -- | -- | 13.45 | 13.45 | 6 | 18 | |
| VT [10-11] | 7.95 | 8.1 | 8.1 | 16.2 | 30.0 | 37.5 | 40.0 | 2 | 16 | 2 | 16 | 2 | 16 | 2 | 16 | -- | -- | 13.5 | 13.5 | 6 | 18 | |
| VT [12-13] | 7.99 | 12.4 | 12.4 | 24.8 | 40.0 | 37.5 | 40.0 | 3 | 16 | 3 | 16 | 3 | 16 | 3 | 16 | -- | -- | 14.6 | 14.6 | 6 | 18 | |
| VT [13-14] | 7.95 | 4.3 | 4.3 | 8.6 | 20.0 | 30.0 | 32.5 | 2 | 12 | 2 | 12 | 2 | 12 | 2 | 12 | -- | -- | 13.3 | 13.3 | 6 | 18 | |
| (VRC) | *1 | 1.5 | 1.5 | 3.0 | 15.0 | 30.0 | 32.5 | 2 | 10 | 2 | 10 | 2 | 10 | 2 | 10 | -- | -- | 6.15 | 6.15 | 6 | 18 | *1 Luz Variable (NIVEL -1) VIGAS RIOSTRAS CENTRADAS |
| (VRM) | *1 | 1.45 | 1.45 | 2.9 | 15.0 | 30.0 | 32.5 | 2 | 10 | 2 | 10 | 2 | 10 | 2 | 10 | -- | -- | 5.95 | 5.95 | 6 | 18 | *1 Luz Variable (NIVEL -1') VIGAS RIOSTRAS MEDIANERAS |
| (VET) | *1 | 0.3 | 0.3 | 0.6 | 15.0 | 20.0 | 22.5 | 2 | 8 | 2 | 8 | 2 | 8 | 2 | 8 | -- | -- | 1.78 | 1.78 | 4.2 | 18 | *1 Luz Variable (NIVEL -1') VIGAS DE ENCADEN. TAPIA FONDO |

# PLANILLA DE VIGAS

## TRAMOS

$\sigma'_{bk}$ = 170 Kg / cm2
$\sigma_{ek}$ = 4200 Kg / cm2

| Nivel | Design. | Luz (m) | P Tn | Q T/m | RA Tm | RB Tm | M(+) tm | bo | do | h | b | Fet cm2 | nº | Ø | Fet' cm2 | nº | Ø | τA kg/cm2 | τB | Estribos Ø | Sep cm | Distrib. | Observaciones |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ① DIR -X- | V 1-2 | 3.55 | ---- | 0.525 | 1.25 | 1.25 | 1.75 | 8.5 | 71.0 | 68.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 2.20 | 2.20 | 6 | 10 | UNIF (*) | ID: V 2-3, V 3-4, V 4-5 "VIGA DINTEL FACHADA" — NOTA: (*) A 15% DE LUZ EN CADA APOYO DENSIFICACION DE ESTRIBOS CERRADOS EN 35% |
| | V 12-13 | 7.95 | 1.72/1.72 | 0.55/1.10 | 6.95 | 7.05 | 19.65 | 90.0 | 31.0 | 28.0 | ---- | 40.0 | 20 | 16 | 8.0 | 4 | 16 | 2.78 | 2.81 | 6 | 17 | UNIF (*') DOBL. | ID: V 13-14 |
| | (Vv1) | (1.20) | 1.50 | 0.75 | ---- | 2.15 | (2.00) | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 6.0 | 3 | 16 | ---- | 2.08 | 6 | 8.5 | UNIF (*) | ID: (Vv3) |
| | VRESC | 4.50 | ---- | 1.40 | 3.0 | 3.0 | 1.70 | 35.0 | 14.0 | 11.0 | ---- | 5.6 | 5 | 12 | 3.1 | 4 | 10 | 2.2 | 2.2 | 6 | 15 | UNIF (*) | |
| | V 15-16 | 3.04 | ---- | 2.10 | 2.32 | 2.32 | 1.48 | 15.0 | 41.0 | 38.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 4.07 | 4.07 | 6 | 15 | UNIF (*) | |
| | V 16-17 | 3.55 | ---- | 0.75 | 0.83 | 0.83 | 0.70 | 15.0 | 41.0 | 38.0 | ---- | 2.4 | 3 | 10 | 2.4 | 3 | 10 | 1.46 | 1.46 | 6 | 15 | UNIF (*) | ID: V 17-18, V 18-19 |
| ① DIR -Y- | (Vv6) | (2.16) | 1.50 | 0.75 | ---- | 2.15 | (2.00) | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 6.0 | 3 | 16 | ---- | 2.08 | 6 | 8.5 | UNIF (*) | |
| | V 3-7 | 2.01 | ---- | 4.75 | 8.90 | 8.90 | 7.85 | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 2.4 | 3 | 10 | 8.6 | 8.6 | 6 | 15 | UNIF (*) | |
| | V 7-10 | 1.63 | ---- | 4.65 | 5.24 | 5.24 | 2.93 | 8.5 | 71.0 | 68.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 8.93 | 8.93 | 6 | 8.5 | UNIF (*) | ID: V 10-13 |
| | V 13-17 | 3.06 | ---- | 4.65 | 5.24 | 5.24 | 2.93 | 8.5 | 71.0 | 68.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 8.93 | 8.93 | 6 | 8.5 | UNIF (*) | ID: V 14-19 |
| | V 5-8 | 2.01 | ---- | 4.55 | 8.6 | 8.6 | 7.81 | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 2.4 | 3 | 10 | 8.31 | 8.31 | 6 | 15 | UNIF (*) | |
| | V 8-11 | 1.63 | ---- | 1.35 | 1.85 | 1.85 | 1.40 | 15.0 | 71.0 | 68.0 | ---- | 2.4 | 3 | 12 | 2.4 | 3 | 10 | 1.79 | 1.79 | 6 | 15 | UNIF (*) | ID: V 11-14 |
| | (Vv T1) | (0.90) | 1.50 | 0.25 | ---- | 2.0 | (1.05) | 15.0 | 26.0 | 23.0 | ---- | 4.5 | 4 | 12 | 4.5 | 4 | 12 | ---- | 5.2 | 6 | 10 | UNIF (*) | |
| | V 6-9 | 1.67 | ---- | 1.35 | 1.72 | 1.72 | 1.45 | 15.0 | 41.0 | 38.0 | ---- | 4.5 | 4 | 12 | 4.5 | 4 | 12 | 3.01 | 3.01 | 6 | 15 | UNIF (*) | ID: V 9-12 |
| | V 12-15 | 3.12 | ---- | 1.00 | 0.8 | 0.8 | 0.75 | 15.0 | 41.0 | 38.0 | ---- | 2.4 | 3 | 10 | 2.4 | 3 | 10 | 1.40 | 1.40 | 6 | 15 | UNIF (*) | |
| | V T1-11 | 2.92 | ---- | 1.00 | 0.8 | 0.8 | 0.75 | 15.0 | 41.0 | 38.0 | ---- | 2.4 | 3 | 10 | 2.4 | 3 | 10 | 1.40 | 1.40 | 6 | 15 | UNIF (*) | ID: V T2-12 |
| ①' DIR -X- | V 15-16 | 3.04 | ---- | 0.725 | 0.96 | 0.96 | 0.90 | 15.0 | 38.5 | 35.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 1.81 | 1.81 | 6 | 15 | UNIF (*) | NOTA: (*) A 15% DE LUZ EN CADA APOYO DENSIFICACION DE ESTRIBOS CERRADOS EN 35% |
| | V 16-17 | 3.55 | ---- | 0.50 | 0.48 | 0.48 | 0.72 | 15.0 | 38.5 | 35.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 0.91 | 0.91 | 6 | 15 | UNIF (*) | ID: V 17-18, V 18-19 |
| | V T2-T1 | 3.14 | ---- | 0.785 | 1.05 | 1.05 | 0.79 | 12.0 | 38.5 | 39.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 2.47 | 2.47 | 6 | 12 | UNIF (*) | |
| | V T2-14/19 | 1.51 | ---- | 0.625 | 0.52 | 0.52 | 0.53 | 12.0 | 38.5 | 35.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 1.22 | 1.22 | 6 | 12 | UNIF (*) | |
| | V 12/15-T1 | 2.67 | ---- | 0.625 | 0.52 | 0.52 | 0.53 | 12.0 | 38.5 | 35.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 1.22 | 1.22 | 6 | 12 | UNIF (*) | |
| ①' DIR -Y- | V 12-15 | 3.12 | 0.96 | 0.725 | 1.85 | 1.10 | 1.20 | 15.0 | 38.5 | 35.5 | ---- | 3.8 | 2/2 | 10/12 | 1.5 | 3 | 8 | 3.47 | 2.07 | 6 | 15 | UNIF (*) | |
| | VT1-15/16 | 2.92 | 0.48 | 0.725 | 1.55 | 1.10 | 1.20 | 15.0 | 38.5 | 35.5 | ---- | 3.8 | 2/2 | 10/12 | 1.5 | 3 | 8 | 2.92 | 2.07 | 6 | 15 | UNIF (*) | ID: V T1-17/18, V T2-12/19 |
| | V T2-16/17 | 2.92 | ---- | 0.625 | 0.52 | 0.52 | 0.53 | 12.0 | 38.5 | 35.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 1.22 | 1.22 | 6 | 12 | UNIF (*) | NOTA: (*) (IDEM) ① y ①' |
| | V 14-19 | 2.92 | ---- | 0.625 | 0.52 | 0.52 | 0.53 | 12.0 | 38.5 | 35.5 | ---- | 3.4 | 3 | 12 | 1.5 | 3 | 8 | 1.22 | 1.22 | 6 | 12 | UNIF (*) | |
| ② DIR -X- | V 1-2 | 3.55 | ---- | 0.525 | 1.25 | 1.25 | 1.75 | 8.5 | 71.0 | 68.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 2.20 | 2.20 | 6 | 10 | UNIF (*) | ID: V 2-3, V 3-4, V 4-5 "VIGA DINTEL FACHADA" — NOTA: (*) A 15% DE LUZ EN CADA APOYO DENSIFICACION DE ESTRIBOS CERRADOS EN 35% |
| | V 12-13 | 7.95 | 1.72/1.72 | 0.55/1.10 | 6.95 | 7.05 | 19.65 | 90.0 | 31.0 | 28.0 | ---- | 40.0 | 20 | 16 | 8.0 | 4 | 16 | 2.78 | 2.81 | 6 | 17 | UNIF (*') DOBL. | ID: V 13-14 |
| | (Vv1) | (1.20) | 1.50 | 0.75 | ---- | 2.15 | (2.00) | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 6.0 | 3 | 16 | ---- | 2.08 | 6 | 8.5 | UNIF (*) | ID: (Vv3) |
| | VRESC | 4.50 | ---- | 1.40 | 3.0 | 3.0 | 1.70 | 35.0 | 14.0 | 11.0 | ---- | 5.6 | 5 | 12 | 3.1 | 4 | 10 | 2.2 | 2.2 | 6 | 15 | UNIF (*) | |
| | V 15-16 | 3.04 | ---- | 2.10 | 2.32 | 2.32 | 1.48 | 15.0 | 41.0 | 38.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 4.07 | 4.07 | 6 | 15 | UNIF (*) | |
| | V 16-17 | 3.55 | ---- | 0.75 | 0.83 | 0.83 | 0.70 | 15.0 | 41.0 | 38.0 | ---- | 2.4 | 3 | 10 | 2.4 | 3 | 10 | 1.46 | 1.46 | 6 | 15 | UNIF (*) | ID: V 17-18 |
| ② DIR -Y- | (Vv6) | (2.16) | 1.50 | 0.75 | ---- | 2.15 | (2.00) | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 6.0 | 3 | 16 | ---- | 2.08 | 6 | 8.5 | UNIF (*) | |
| | V 3-7 | 2.01 | ---- | 4.75 | 8.90 | 8.90 | 7.85 | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 2.4 | 3 | 10 | 8.6 | 8.6 | 6 | 15 | UNIF (*) | |
| | V 7-10 | 1.63 | ---- | 4.65 | 5.24 | 5.24 | 2.93 | 8.5 | 71.0 | 68.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 8.93 | 8.93 | 6 | 8.5 | UNIF (*) | ID: V 10-13 |
| | V 13-17 | 3.06 | ---- | 4.65 | 5.24 | 5.24 | 2.93 | 8.5 | 71.0 | 68.0 | ---- | 4.5 | 4 | 12 | 2.4 | 3 | 10 | 8.93 | 8.93 | 6 | 8.5 | UNIF (*) | ID: V 14-19 |
| | V 5-8 | 2.01 | ---- | 4.55 | 8.6 | 8.6 | 7.81 | 15.0 | 71.0 | 68.0 | ---- | 6.2 | 2/2 | 12/16 | 2.4 | 3 | 10 | 8.31 | 8.31 | 6 | 15 | UNIF (*) | |
| | V 8-11 | 1.63 | ---- | 1.35 | 1.85 | 1.85 | 1.40 | 15.0 | 71.0 | 68.0 | ---- | 2.4 | 3 | 12 | 2.4 | 3 | 10 | 1.79 | 1.79 | 6 | 15 | UNIF (*) | ID: V 11-14 |
| | (Vv T1) | (0.90) | 1.50 | 0.25 | ---- | 2.0 | (1.05) | 15.0 | 26.0 | 23.0 | ---- | 4.5 | 4 | 12 | 4.5 | 4 | 12 | ---- | 5.2 | 6 | 10 | UNIF (*) | |
| | V 6-9 | 1.67 | ---- | 1.35 | 1.72 | 1.72 | 1.45 | 15.0 | 41.0 | 38.0 | ---- | 4.5 | 4 | 12 | 4.5 | 4 | 12 | 3.01 | 3.01 | 6 | 15 | UNIF (*) | ID: V 9-12 |
| | V 12-15 | 3.12 | ---- | 1.00 | 0.8 | 0.8 | 0.75 | 15.0 | 41.0 | 38.0 | ---- | 2.4 | 3 | 10 | 2.4 | 3 | 10 | 1.40 | 1.40 | 6 | 15 | UNIF (*) | |
| | V T1-11 | 2.92 | ---- | 1.00 | 0.8 | 0.8 | 0.75 | 15.0 | 41.0 | 38.0 | ---- | 2.4 | 3 | 10 | 2.4 | 3 | 10 | 1.40 | 1.40 | 6 | 15 | UNIF (*) | ID: V T2-12 |
| ③ | VETM | *1 | ---- | 0.15 | 0.3 | 0.3 | 0.125 | 15.0 | 15.0 | 12.0 | ---- | 1.00 | 2 | 8 | 1.00 | 2 | 8 | 1.00 | 1.00 | 4.2 | 18 | UNIF (*) | (DIR -X-e-Y-) *1 LUZ VARIABLE "VIGA DE ENCADENADO SUPERIOR" — NOTA: (*) (IDEM) ① y ①' |

NOTA: PANILLA RESUMEN DE ACCIONES GRAVITATORIAS Y DINAMICAS.-

# PLANILLA DE COLUMNAS

FORMAS ① ② ③ ④ ⑤

$\sigma'_{bk}$ = 170 Kg / cm2
$\sigma_{ek}$ = 4200 Kg / cm2
ACERO TIPO = ADN (420
COEF. DE SEGUR. (γ) = 2.1

| Nivel | Desig. | hp m | N tn | d1 cm | d2 cm | s cm2 | Nº | Ø | secc. cm2 | Estribos Ø | Sep. cm | λ | W | Wpv | OBSERVACIONES | | | FORMA |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (-2/0/1) | C 1 | 3.40 | 6.50 | 20 | 20 | 400 | 4 | 12 | 4.5 | 6 | 12.0 | 17.0 | 1.05 | 14.33 | ID: C04 | VIVIENDA | EST. CERR. SIMPLE | ④ |
| (-2/0/1) | C 2 | 3.40 | 12.70 | 20 | 25 | 500 | 4 | 12 | 4.5 | 6 | 12.0 | 17.0 | 1.05 | 28.00 | ID: C03 | VIVIENDA | EST. CERR. SIMPLE | ④ |
| (-2/0/1) | C 3 | 3.20 | 11.65 | 25 | 18 | 450 | 6 | 12 | 6.8 | 6 | 12.0 | 17.8 | 1.06 | 26.90 | ID: Cc2-Cc5-Cc6 | COCHERA | EST. CERR. SIMPLE Y GANCHO CERRADO | ③ |
| (-2/0/1) | C 4 | 3.20 | 14.5 | 20 | 20 | 400 | 6 | 12 | 6.8 | 6 | 12.0 | 16.0 | 1.02 | 31.06 | ID: Cc4'-Cc4'' | COCHERA | EST. CERR. SIMPLE Y GANCHO CERRADO | ④ |
| (-2/0/1) | CE1 | 3.20 | 13.4 | 40 | 40 | 1600 | 4 | 12 | 7.6 | 6 | 12.0 | 8.0 | 1.00 | 28.14 | ID: Cc7 | COCHERA | EST. CERR. SIMPLE Y GANCHO CERRADO DOBLES | ④ |
| (-2/0/1) | C 1 | 3.40 | 17.4 | 20 | 25 | 500 | 6 | 12 | 6.8 | 6 | 12.0 | 17.0 | 1.05 | 38.37 | ID: C4-C10-C20-C23-C28 | VIVIENDA | EST. CERR. SIMPLE Y GANCHO CERRADO | ② |
| (1/2) | C 2 | 3.00 | 7.2 | 20 | 25 | 500 | 4/2 | 12/10 | 6.1 | 6 | 12.5 | 15.0 | 1.00 | 19.32 | ID: C4-C10-C20-C23-C28 | VIVIENDA | EST. CERR. SIMPLE Y GANCHO CERRADO | ② |
| (2/3) | C 3 | 1.20 | 5.0 | 20 | 25 | 500 | 4/2 | 12/10 | 5.5 | 6 | 13.0 | 8.0 | 1.00 | 12.38 | | T.RESERVA | EST. CERR. SIMPLE Y GANCHO CERRADO | ② |
| (3/4) | C 4 | 7.40 | 2.4 | 20 | 25 | 500 | 4/2 | 10/8 | 2.6 | 6 | 15.0 | 12.0 | 1.00 | 5.04 | | T.RESERVA | EST. CERR. SIMPLE Y GANCHO CERRADO | ② |
| (-2/0/1) | C 1 | 3.40 | 28.5 | 30 | 18 | 540 | 4/2 | 12/16 | 7.6 | 6 | 12.0 | 18.0 | 1.06 | 52.21 | ID: C7-C21-C24 | VIVIENDA | EST. CERR. DOBLE | ③ |
| (1/2) | C 2 | 3.00 | 12.2 | 30 | 18 | 540 | 4/2 | 12/10 | 6.5 | 6 | 12.5 | 16.7 | 1.06 | 26.90 | ID: C7-C21-C24 | VIVIENDA | EST. CERR. DOBLE | ③ |
| (2/2') | C 3 | 1.00 | 4.5 | 30 | 18 | 540 | 4/2 | 10/8 | 4.2 | 6 | 13.0 | 5.55 | 1.00 | 9.45 | | VIVIENDA | EST. CERR. DOBLE | ③ |
| (2/3) | C 4 | 1.20 | 7.3 | 30 | 18 | 540 | 4/2 | 12/10 | 6.5 | 6 | 13.0 | 6.67 | 1.00 | 15.33 | | T.RESERVA | EST. CERR. DOBLE | ③ |
| (3/4) | CE1 | 2.40 | 2.2 | 30 | 18 | 540 | 4/2 | 8/6 | 3.2 | 6 | 15.0 | 13.3 | 1.00 | 4.62 | | T.RESERVA | EST. CERR. DOBLE | ③ |
| (-1/0/1) | (CE1) | 3.00 | 2.25 | 18 | *1 | ---- | 4 | 10 | 3.1 | 6 | 15 | 16.7 | 1.05 | 4.96 | ID: (CEM) *1: ANCHO DE MURO | VIVIENDA Y COCHERA | EST. CERR. SIMPLE | ⑤ |
| (-1/0/1) | (CE) | 3.00 | 1.62 | 18 | *1 | ---- | 4 | 10 | 3.1 | 6 | 15 | 16.7 | 1.05 | 3.13 | *1: ANCHO DE MURO | VIVIENDA | EST. CERR. SIMPLE | ⑤ |
| (-1/0/1) | (CE) | 3.00 | 1.25 | 12 | *1 | ---- | 4 | 10 | 3.1 | 6 | 15 | 25.0 | 1.33 | 3.49 | *1: ANCHO DE MURO | COCHERA | EST. CERR. SIMPLE | ⑤ |
| (-1/0/1) | (CE') | 3.00 | 1.53 | 12 | *1 | ---- | 4 | 10 | 3.1 | 6 | 15 | 25.0 | 1.33 | 4.28 | *1: ANCHO DE MURO (IDEM PLANTA ALTA) | VIVIENDA | EST. CERR. SIMPLE | ⑤ |

NOTA: PANILLA RESUMEN DE ACCIONES GRAVITATORIAS Y DINAMICAS.-

# PLANILLA DE BASES DE HORMIGON ARMADO

EXCENTRICAS — CENTRADAS

$\sigma'_{bk}$ = 170 Kg / cm2
$\sigma_{ek}$ = 4.200 Kg / cm2
$\sigma_t$ adm = 0.850 Kg / cm2

| DESIG. | N Tn | S m2 | ax | ay | d | D | h | M (Tm) | Ax cm2 | Nº | Ø | Ay cm2 | Nº | Ø | σc Kg/cm2 | σp Kg/cm2 | observaciones |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B/1 | 24.50 | 1.96 | 1.40 | 1.40 | 0.235 | 0.925 | 0.875 | 16.8 / 17.25 | 9.00 | 18 | 8 | 9.00 | 18 | 8 | 3.25 | 3.44 ID: B/C3 | EXCENTRICA ESQUINIL. MUNIC. |
| B/2 | 11.50 | 2.20 | 2.00 | 1.10 | 0.15 | 0.60 | 0.55 | 2.00 / 2.88 | 4.50 | 9 | 8 | 8.50 | 17 | 8 | 2.11 | 2.19 ID: B/C2/C3/C4 | EXCENTRICA L. MUNICIPAL |
| B/3 | 28.75 | 2.40 | 1.20 | 2.00 | 0.225 | 0.875 | 0.825 | 11.4 / 6.20 | 13.0 | 26 | 8 | 6.50 | 13 | 8 | 3.30 | 3.47 ID: B/C8 | EXCENTRICAS |

NOTA: Profundidad de Excavación - Fundaciones ( Df= - 1.75 m ) (NIVEL -2); NIVEL INTERIOR DE REFERENCIA = +0.14 m

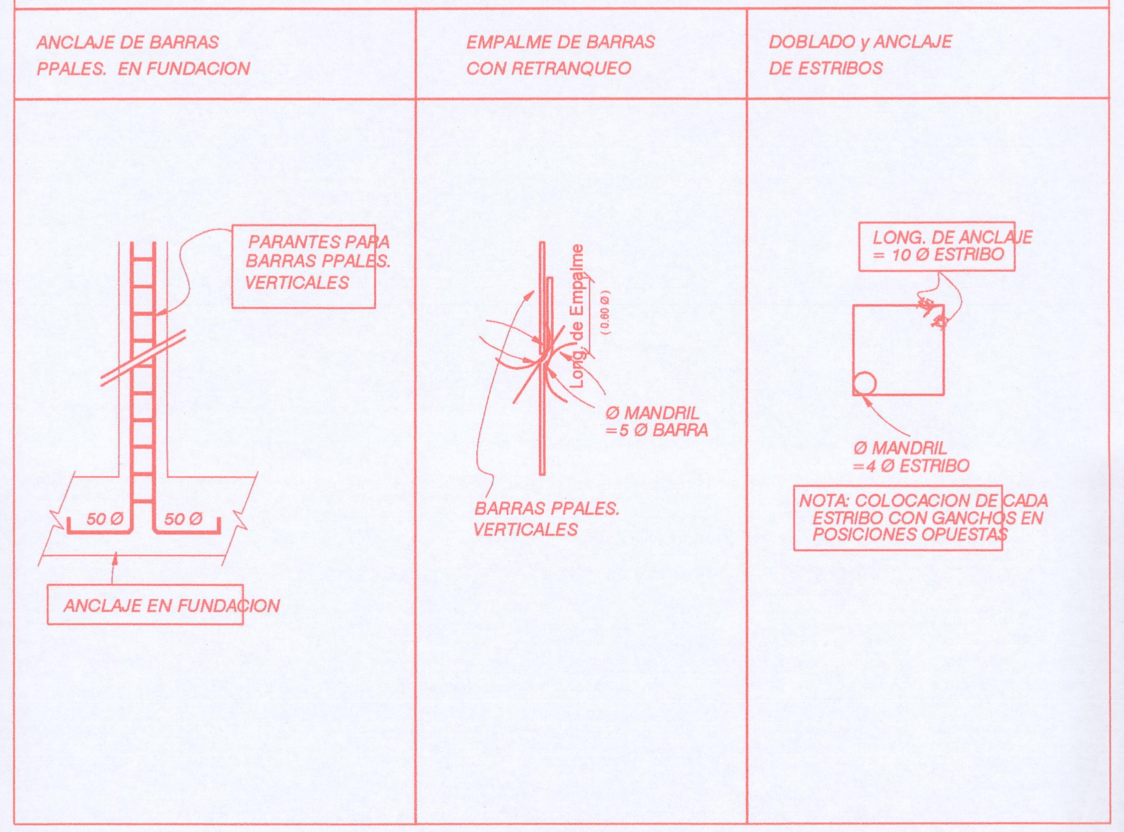
# PLANILLA DE CIMENTACION TRADICIONAL

HORMIGON SIMPLE (H13) — CENTRADAS — EXCENTRICAS

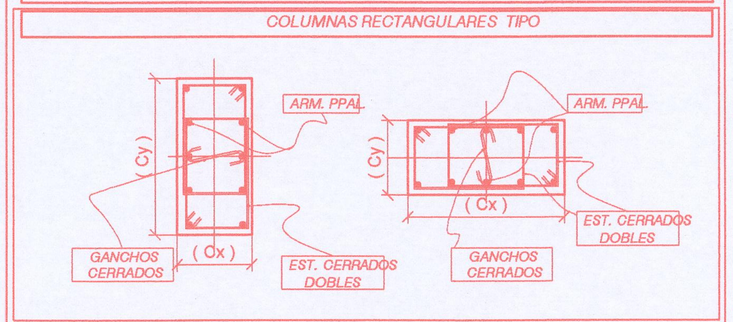
| design. | N Tn | S m2 | H | DIMENSIONES m | L m | OBSERVACIONES |
|---|---|---|---|---|---|---|
| Z | 2.15 | 0.43 | 0.35 | | Variab. | Excentrico Edificación |

NOTA: Profundidad de Excavación / Cimentación ( Hc= - 0.35 m ) (NIVEL -1) De Nivel Interior de Referencia= +0.14m

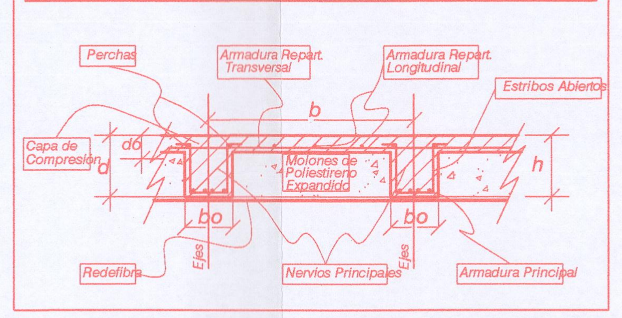
# DETALLES : ANCLAJES EMPALMES Y DOBLADOS ARMADURAS PRINCIPALES VERTICALES Y ESTRIBOS



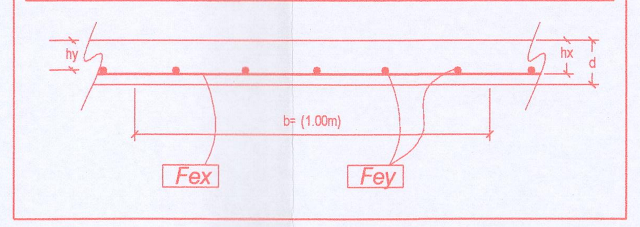

COLUMNAS : ARMADURAS PRINCIPALES, ESTRIBOS, ANCLAJES, DOBLADOS Y EMPALMES
COLUMNAS RECTANGULARES TIPO



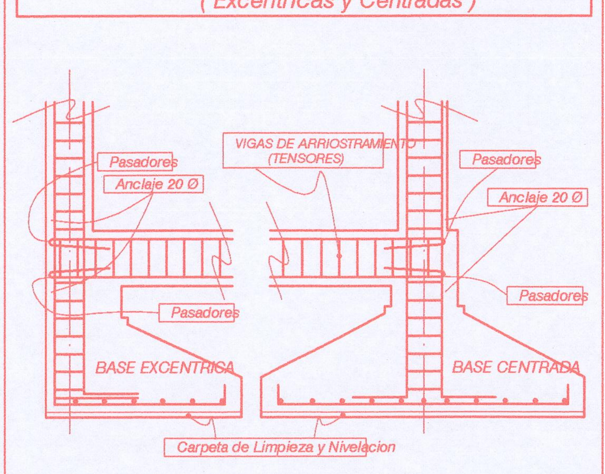

## LOSAS NERVURADAS (Referencias)



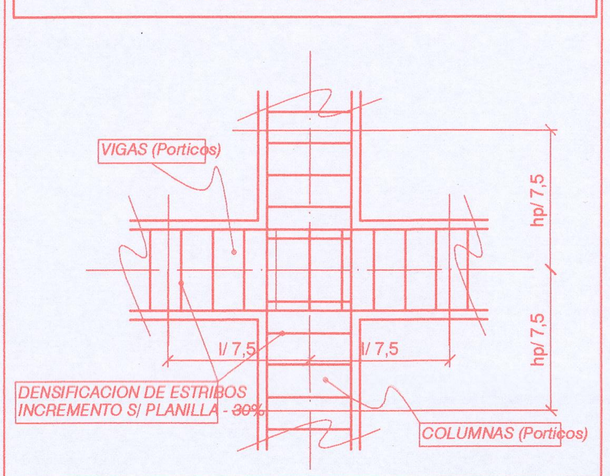

## LOSAS MACIZAS (Referencias)




## VIGAS TENSORES EN FUNDACION ( Excentricas y Centradas )

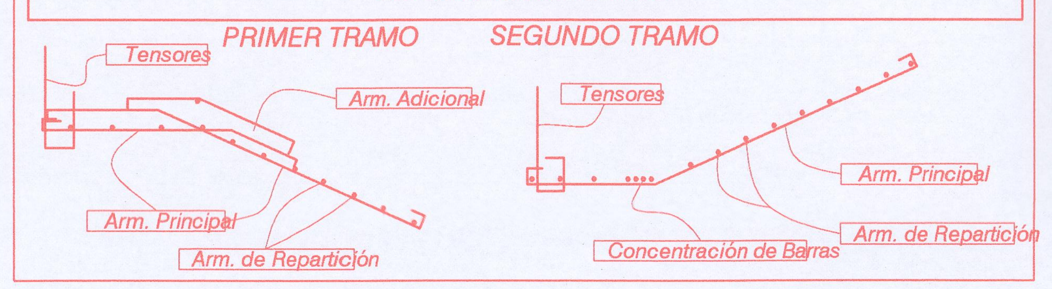



## NUDOS DE PORTICOS




## LOSAS ESCALERAS

PRIMER TRAMO — SEGUNDO TRAMO

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DIST.: 02 ZONA.: 06 MANZ.: 060 PARC.: 132

PADRON INMOB.: 01-1-147689 I.C.M.: 607-T-2021 DOMINIO: 7S-35.914

LEY 4683 211611

DEST. DE LA OBRA : VIVIENDA FAMILIAR

PROPIETARIOS: TSANG, SEBASTIAN D.N.I.: 26.398.057 CUIL 20-26398057-4
CORVERA, MARIA VALERIA D.N.I.: 26.774.610 CUIL 27-26774610-7

UBICACION: AV. LOS LAPACHOS S/Nº. CALLE PRIVADA DE PASO COMÚN S/Nº - LOTE 32

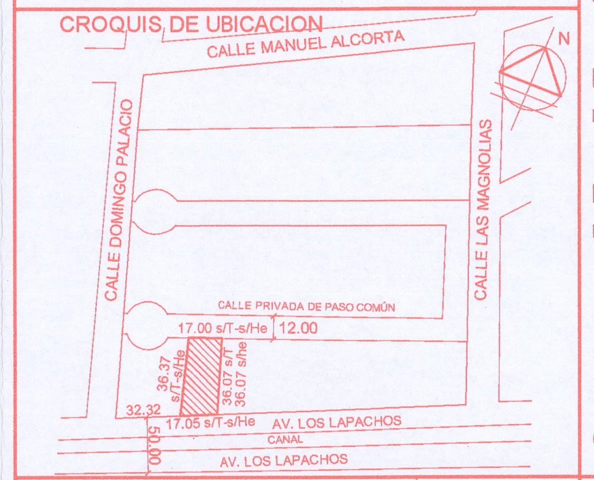
BARRIO: ALMIRANTE BROWN

CIUDAD: SANTIAGO DEL ESTERO

ESCALA: 1 : 100 - 1 : 200

| | |
|---|---|
| SUP. DE TERRENO: | 607,26m2 s/T -s/he |
| SUP. CUBIERTA PLANTA BAJA | 212,88 m2 |
| SUP. CUBIERTA PLANTA ALTA | 119,35 m2 |
| SUP. CUBIERTA TOTAL : | 332,03 m2 |

CROQUIS DE UBICACION




PROPIETARIO: DOMICILIO:
PROYECTO : DOMICILIO:
DIRECCION TECNICA : DOMICILIO:
ELECTRICIDAD : DOMICILIO:
CATEG. DEL COLEGIO DE ARQUITECTOS: "B"

| | | C. PRIV. COM. | AV. LOS LAPACHOS |
|---|---|---|---|
| ANCHO DE CALLE : | 12.00 m | 50.00 m | |
| ANCHO DE VEREDA : | 3.00 m | 3.00 m | |
| ANCHO DE CALZADA : | 6.00 m | 4.00 m | |

| | C. PRIV. COM. | AV. LOS LAPACHOS |
|---|---|---|
| PAVIMENTO : | SI | NO |
| SERVICIOS CLOACALES : | NO | NO |

DESIGNACION : PLANO DE ESTRUCTURAS ⑥

OBSERVACIONES: Las referencias al Título de dominio son meramente indicativas, y la aprobación administrativa no valora los derechos que de ellos emergen.


12 ENE. 2022
MUNICIPALIDAD DE LA CIUDAD DE SANTIAGO DEL ESTERO — SECRETARIA DE ECONOMIA — APROBADO — DIRECCION DE SUELO URBANO — EXPEDIENTE: 11 - T - 2021 — FECHA: 21 MAR 2022
DIRECCION DE SUELO URBANO — PERMISO DE COMIENZO DE OBRA Nº 4672
DIRECCION DE SUELO URBANO — EXPEDIENTE 64301 — FECHA DE SELLADO 29.11.21 — BOLETA Nº
SECRETARIA DE ECONOMIA — Municipalidad de la Capital — DIREC. DE SUELO URBANO
PROYECTO Y DIRECCION DE OBRA — 2 1 NOV 2021
Arq. Eva Luz Khairallah — SECRETARIA TECNICA — COLEGIO DE ARQUITECTOS S.D.E.
Ing. Agrim. JUAN JOSE FRIAS — DIRECTOR — DIRECCION DE CATASTRO — MUNICIPALIDAD DE SGO. DEL ESTERO
Arq. Bravo, Ma. Esther — Dpto. OBRAS PRIVADAS — DIRECCION DE SUELO URBANO
Arq. BERNARDO EMIL SANTILLAN — DIRECTOR — DIRECCION DE SUELO URBANO — MUNICIPALIDAD DE SGO. DEL ESTERO